Seat: TBD PETER ANTEATER PANTEATE

CompSci 161 Diagnostic Exam 2-2, Fall 2025

DO NOT OPEN EXAM UNTIL INSTRUCTED TO DO SO

SILENCE AND STORE ALL ELECTRONICS

This is a diagnostic exam intended to help you evaluate your readiness for the real exam.

The following rules apply to you, whether you think they do or not. Read and understand them; failure to abide by these rules, or directions given by course staff during the exam, may result in disciplinary action, including but not limited to a failing grade in the class.

- This exam is solely for students enrolled in this lecture. Anyone not enrolled in this lecture may not take an exam.
- You may not open the exam or begin writing until the instructor has explicitly given you permission to do so.
- Keep acceptable identification readily accessible during the test. Proctors may request to see it.
- This exam is closed book, closed notes, and is individual effort. Once course staff begin passing out exams, you may not communicate with anyone other than proctors for any reason, nor may you have electronics, including calculators watches and phones, available to you during the test for any reason. **YOU DO NOT NEED A CALCULATOR!**
- If you leave your seat during the test for any reason, your instructor may collect it and deem you to have turned it in. Do not ask proctors for an exemption to this, they are not authorized to grant such.
- You must take the exam in your assigned seat unless the professor (not a TA) tells you otherwise. The instructor will call to cease writing at 6:45 PM, at which point you must immediately cease writing and close the exam. You may not write any further at that point, including finishing one's current sentence.
- If you believe a question is ambiguous, write at least two reasonable interpretations and indicate clearly which one you will be using. Then answer your question with that assumption. Unless your interpretation makes the problem much more trivial than intended, we will grade your response as if one of us had made that clarification.
- The purpose of the real exam is to evaluate how well *you* understand the material presented in the course. It is an academic integrity violation to do anything that subverts the goals of this assessment including, but not limited to, not doing your own work or submitting that of anyone else.
- We will only grade responses marked in the space provided for each question.

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Nothing you write on this page will be graded. The next page in this booklet contains a spot to answer these questions. You may use this page as scratch paper if you would like, and room to do so exists.

1. (1.5 point) Consider the following table, which is the output from running the optimal binary search tree algorithm for some seven-key input. In each entry, the value printed in the upper half of the cell is Tree$[i, j]$, the cost of the optimal binary search tree consisting of keys $i \ldots j$. The value printed in the lower half of the cell is the value of roots$[i, j]$.

   Suppose we have computed the following using the algorithm for the optimal binary search tree. However, we accidentally forgot to record Tree$[1, n]$. What is the missing value for that spot? It is the only one omitted in the following table.

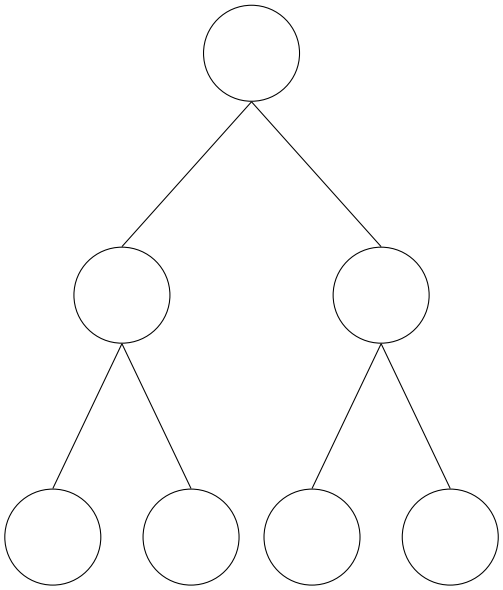
	k_1	k_2	k_3	k_4	k_5	k_6	k_7	k_8	k_9	k_{10}
k_1	0.12 1	0.3 1	0.55 2	0.73 2	1.12 3	1.51 3	1.8 5	2.17 5	2.45 5	 5
k_2		0.09 2	0.29 3	0.41 3	0.79 3	1.09 5	1.36 5	1.73 5	2.01 5	2.25 5
k_3			0.11 3	0.23 3	0.56 5	0.82 5	1.09 5	1.46 5	1.72 6	1.96 6
k_4				0.06 4	0.28 5	0.54 5	0.81 5	1.12 6	1.33 6	1.57 6
k_5					0.16 5	0.42 5	0.63 6	0.94 6	1.15 6	1.39 6
k_6						0.13 6	0.31 6	0.57 7	0.78 7	0.96 8
k_7							0.09 7	0.29 8	0.43 8	0.61 8
k_8								0.11 8	0.25 8	0.41 9
k_9									0.07 9	0.19 9
k_{10}										0.06 10

2. (1.5 point) Draw the root, the root's children, and the root's grandchildren for the optimal binary search tree for the input probabilities in the previous question.

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Write your answer to question 1 in this box.

Write the keys for question 2 into this tree. If a subtree is null, leave the entry blank.



You may use the rest of this page as scratch paper if you wish, but nothing outisde of the designated answer areas will be graded.

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3. (2 points) Suppose you are running a boat that takes cars across the water to Catalina Island (despite that, in real life, cars are not allowed on that island). Your boat is L yards long and three lanes wide. Each lane is wide enough to hold one car. There are currently n cars in line that want to get onto your boat. If there is room on the boat for this, you can direct the car at the front of the line onto one of the lanes on your boat. It will drive onto your boat and as far forward in that lane as it can go. The cars *must* get on the boat in the order they arrived. If all three lanes cannot fit the front car, then no more cars can get onboard the boat.

   Your favorite algorithms professor owns a banana stand at the dock where the cars get on the boat, so you want to allow *as few* cars as possible onto the boat (so the drivers will have to wait for the next boat, and might buy a frozen banana in the meantime). However, you don't want it to be obvious that you are doing this: if the vehicle at the front of the line can fit onto the boat, you must allow it onto the boat (in a lane where it will fit).

   You can see the lengths of all vehicles in the queue on your security camera. **Use dynamic programming** to design an algorithm to decide how many cars get on the boat. The input is an integer L (the length of the boat, in yards), the number n of cars in line, and an array carlen$[1 \ldots n]$ containing the (integer) lengths (in yards) of all vehicles in line. You may assume that, for each i, carlen$[i] \leq L$.

   The output (return value) of your function is the smallest value k such that cars $1 \ldots k$ can be placed (in some manner) on the boat, but car $k+1$ cannot. When you express your running time, do so in terms of L and/or n, as appropriate.

   Example: Suppose $L = 6$ and carlen $= \{3, 3, 4, 4, 2, 2\}$. You would put the first three cars in different lanes, causing the fourth to not fit. For this input, you return 3. Yes, there is a way to make all six fit, but your goal is to make the *fewest* fit.

   Hint: if the first $m-1$ cars (for some m) are on the boat, and you are deciding which lane to put the mth car (if that's even possible), what information is available to you?

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Write your answer to question 3 on this page. It is possible that you will not use the entire page.

Scratch page. Nothing you write on this side of the page will be graded.

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